Probability theory

Avg temp prediction in a month.

2026, June it will rain or not?

It'll check last 100 years data.

Let's say, 90 times it rained.

So, 90% chance that

it will rain in

June 2026.

Terminology:

- Events/Variables

can be named anything. (A, B, C, X, Y, Z)

Each variable/event has a domain.

Lets take X, Y.

Event $X(X, \neg X)$

Event $Y(Y, \neg Y)$

$X \rightarrow$ x will happen

$\neg X \rightarrow$ x " not "

so domain has only two values.

However, unde one variable there can be multiple domain value.

Lets say weather.

Its not only sunny not sunny.

$W \to$ (sunny, cloudy, snowy,
Rainy, Foggy, Misty)

Each value will have a probability associated with it.
And summation of all the probabilities of the domain values will be 1.

W	$P(W)$
Sunny	0.25
Cloudy	0.25
Snowy	0.25
Rainy	0.15

$P(\text{Rainy}) = 0.15$

Foggy	0.05
Misty	0.05

$= 1$

$P(X)$, $P(\neg X)$ previously seen events.

$$\boxed{0 \leq P(\text{event}) \leq 1}$$

$$\boxed{P(A),\quad P(\neg A) = 1 - P(A)}$$

$$P(A) + P(\neg A) = 1$$

$$\Rightarrow P(A) = 1 - P(\neg A)$$

or $P(\neg A) = 1 - P(A)$

Probability can be a combination of multiple values.

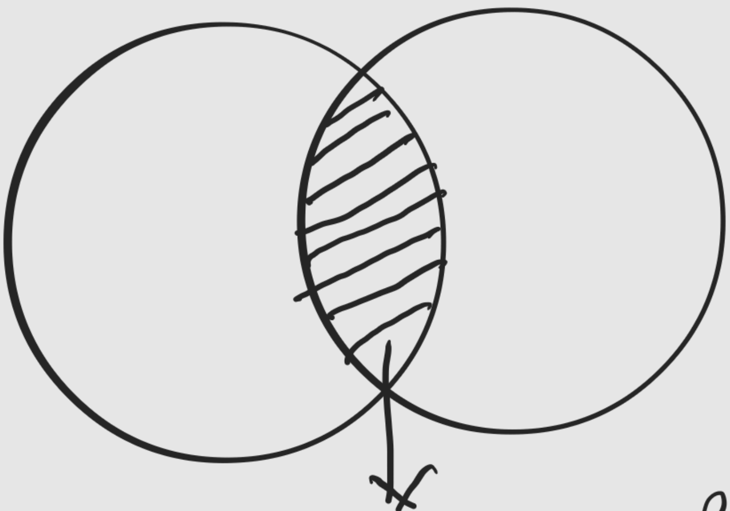
$P(A \cap B) \rightarrow$ Joint probability

or, $P(A \text{ and } B)$

$P(A, B)$

It means A & B happening together

Venn Diagram :



A & B ; $P(A \cap B)$

$P(A \vee B) \rightarrow$ It means $P(A), P(B)$

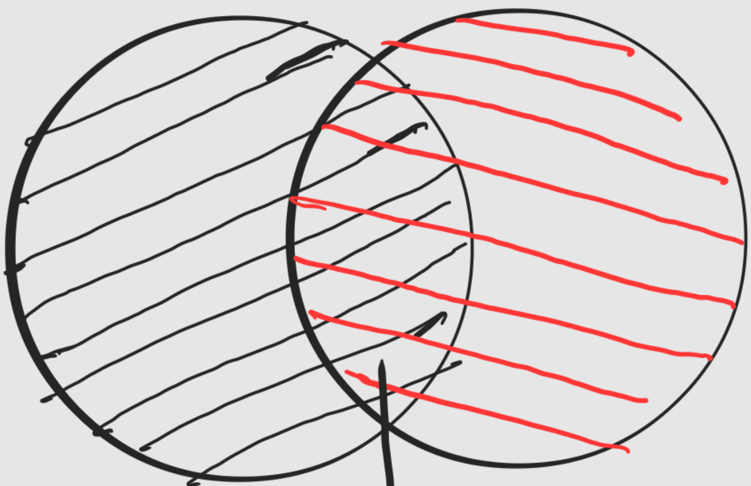
$P(A \text{ or } B)$ on $P(A \text{ and } B)$

#2)

$P(A \vee B) = \underline{P(A)} + \underline{P(B)} - P(A \wedge B)$

venn:

$P(A)$ $P(B)$



we see $P(A \wedge B)$ 2 times,

hence, we deduct 1 of it.

Conditional Probability

#) $P(A|B) = \frac{P(A \cap B)}{P(B)}$

A given B. Meaning B has already occured, now find what's the probability of A happening.

For example, Your class is at 10:00 AM. Your usual time to reach the uni is 20 mins. You missed the bus at 09:30 find what's the probability you will reach the uni on time?

$P(\text{on time} \mid \text{missed bus})$?

Joint Probability:

$P(A \cap B)$

Probability of A and B happening at the same time.

let's say,

$A \longrightarrow A, \neg A$

$B \longrightarrow B, \neg B$

$$\left.\begin{array}{l} P(A \wedge B) \\ P(A \wedge \neg B) \\ P(\neg A \wedge B) \\ P(\neg A \wedge \neg B) \end{array}\right\}\ \text{Joint Probability distribution}$$

We can represent it using a table.

	A	$\neg A$
B	$P(A \wedge B)$	$P(\neg A \wedge B)$
$\neg B$	$P(A \wedge \neg B)$	$P(\neg A \wedge \neg B)$

fig: Joint distribution table

example with values

	A	$\neg A$
B	0.25	0.50
$\neg B$	0.12	0.13

$$P(A \wedge B) + P(A \wedge \neg B) + P(\neg A \wedge B) + P(\neg A \wedge \neg B) = 1$$

Let's take a peek at the slides example.

alarm $\rightarrow a, \neg a$

burglary $\rightarrow b, \neg b$

	a	$\neg a$
b	0.09	0.01
$\neg b$	0.1	0.8

$p(a \wedge b) = 0.09$

$p(a \wedge \neg b) = 0.1$

$p(b \wedge \neg a) = 0.01$

$p(\neg b \wedge \neg a) = 0.8$

you can find individual events probability as well.

$p(a) = ?$ add alarm column

$p(a) = 0.09 + 0.1$

$= 0.19$

$p(\neg a) = 0.01 + 0.8$

$= 0.81$

Now, $0.09 = ?$

$$0.09 = p(a \wedge b)$$

$0.1 = ?$

$$0.1 = p(a \wedge \neg b)$$

$$p(a) = p(a \wedge b) + p(a \wedge \neg b)$$

$p(b) =$ add the row

$$p(b) = 0.09 + 0.01$$

$$= 0.1$$

$$p(\neg b) = 0.1 + 0.8$$

$$= 0.9$$

$$p(a \mid b) = ?$$

$$P(a \mid b) = \frac{P(a \wedge b)}{P(b)}$$

$$= \frac{0.09}{0.09 + 0.01}$$

$$= 0.9$$

$$P(\neg a \mid \neg b) = ?$$

$$= \frac{P(\neg a \wedge \neg b)}{P(\neg b)}$$

$$= \frac{0.8}{0.1 + 0.8}$$

$$= 0.88$$

Students took exam & cheated on act

	Yes	No	
M	0.32	0.22	0.54
F	0.28	0.18	0.46
	0.60	0.40	

Let's look another example.

Gender & cheating (cont)

$$p(F) = \underbrace{p(F \wedge c)}_{} + \underbrace{p(F \wedge \neg c)}_{}$$

$$= \underbrace{0.28 + 0.18}_{\text{or, add female row}}$$

$$p(F \wedge \neg c) = 0.18$$

$$p(M \mid c) = ?$$

$$= \frac{p(M \wedge c)}{p(c)} = \frac{0.32}{0.60}$$

3 variables:

	t		$\neg t$	
	x	$\neg x$	x	$\neg x$
c	0.108	0.012	0.072	0.008
$\neg c$	0.016	0.064	0.144	0.576

3 variable examples

$t \rightarrow$ toothache (cont.)

$x \rightarrow$ x-ray shows cavity

$c \rightarrow$ cavity

3 variables

2 options for each

So, total outcome $= 2^3$

$= 8$

4 variables

3 options for each

So, total outcome $= 3^4$

$= 81$

what's 0.108?

$= P(t \wedge n \wedge c)$

what's 0.008?

$= P(\neg t \wedge \neg n \wedge c)$

find $P(t \wedge c)$?

$= 0.108 + 0.012$

$= 0.12$

$P(\neg c \wedge n)$?

$= 0.016 + 0.144$

$= 0.16$

$P(\neg n)$?

$= 0.012 + 0.064 + 0.008 + 0.576$

$=$

$=$

$$p(t \mid \neg x) = ?$$

$$= \frac{p(t \wedge \neg x)}{p(\neg x)}$$

$$= \frac{0.012 + 0.064}{0.012 + 0.064 + 0.008 + 0.576}$$